
Learning What to Learn: Hippocampal Memory Consolidation for Continual Model Adaptation

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Abstract

Contemporary large language models are trained on internet-scale corpora and scaled to massive parameter regimes, incurring substantial costs in training, deployment, and continual adaptation. Human cognition, by contrast, is selective and experience-driven, relying on hippocampal mechanisms that preferentially encode salient, novel, and behaviorally relevant experiences and coordinate their gradual consolidation into long-term neocortical memory. We propose **Hippocampus-Augmented Transformer (HAT)**, a memory-centric continual learning mechanism inspired by hippocampal–neocortical consolidation in the mammalian brain. HAT enables language models to learn *what to learn* from interaction by coupling online experience acquisition with offline memory consolidation. During interaction, a base model (the *Cortex*) processes inputs while a *Hippocampus Agent* encodes interactions and feedback into structured memory traces and selectively retains informative experiences based on uncertainty, novelty, and supervision signals, optionally querying an external oracle when needed. Selected experiences are progressively consolidated into a persistent *Neocortex* memory and subsequently replayed during slow-wave sleep (SWS), where the Cortex is updated through parameter-efficient fine-tuning. This wake–sleep separation transforms sparse, interaction-driven feedback into reusable training signals, enabling continual self-improvement without indiscriminate data accumulation. We evaluate HAT on knowledge-intensive question answering, instruction following, and personalization benchmarks, demonstrating consistent improvements over fine-tuning and retrieval-augmented baselines under continual adaptation settings.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across a broad spectrum of natural language tasks [Brown et al., 2020, Touvron et al., 2023, OpenAI, 2023]. Yet a fundamental tension remains: once training concludes, a model’s knowledge is *frozen*, and adapting it to new domains, user preferences, or corrected information requires costly retraining on carefully curated data. This limitation is particularly acute for lightweight, on-device language models in the 0.5–10 B parameter range, where users interact with a personal assistant that must evolve over time but lacks the infrastructure for large-scale retraining.

Existing approaches to post-deployment adaptation fall broadly into two categories. *Retrieval-augmented generation* (RAG) methods [Lewis et al., 2020, Guu et al., 2020] augment the model’s context with external knowledge at inference time, but they do not update the model’s parameters and thus cannot internalise corrections or new skills. *Continual fine-tuning* methods [Kirkpatrick et al., 2017, De Lange et al., 2021] update model weights on new data, but they typically require hand-curated datasets and are susceptible to catastrophic forgetting when data is noisy or unbalanced. Neither paradigm addresses the upstream question: *what should a model learn from?*

In biological intelligence, this problem is solved elegantly. The mammalian hippocampus does not store all sensory experience; instead, it selectively encodes salient events and, during sleep, replays and consolidates them into long-term neocortical memory [McClelland et al., 1995, Kumaran et al., 2016]. This process of *selective memory consolidation* ensures that only high-value experiences—those that are novel, surprising, or corrective—are used to update long-term knowledge.

Inspired by this principle, we propose **HAT (Hippocampus-Augmented Transformer)**, a multi-agent framework that equips a lightweight language model with the ability to *learn what to learn* from its own interactions. HAT introduces a dedicated *Hippocampus Agent* that monitors the base model’s (the *Cortex*) interactions with users, abstracts raw dialogues into structured memory traces, selects high-value experiences using uncertainty, feedback, and novelty signals, and constructs training-ready replay data. During offline phases—analogue to *slow-wave sleep* (SWS) in the brain—curated experiences are replayed to fine-tune the Cortex, producing an improved model that re-enters the interaction loop. When the Cortex is uncertain or produces errors, HAT can also consult an external *Oracle* (a stronger teacher model), enabling a form of on-demand knowledge distillation.

Contributions. Our contributions are three-fold:

1. We formalise the problem of *selective experience curation* for self-improving language models and propose HAT, a memory-centric framework that decouples experience acquisition from parameter optimisation.
2. We design a Hippocampus Agent with three composable stages—*abstraction*, *selection*, and *replay construction*—that convert raw interactions into high-quality training signals using uncertainty estimation, user feedback, and novelty detection.
3. We conduct extensive experiments on knowledge-intensive QA, instruction following, and personalisation benchmarks, demonstrating that HAT achieves consistent improvements in sample efficiency, error-rate reduction, and long-horizon adaptation over strong fine-tuning and retrieval-augmented baselines.

2 Related Work

2.1 Continual Learning for Language Models

Continual learning (CL) aims to update model parameters on a stream of tasks without catastrophic forgetting [Kirkpatrick et al., 2017, Zenke et al., 2017]. Regularisation-based methods penalise changes to important weights [Kirkpatrick et al., 2017, Li and Hoiem, 2017], while replay-based methods store or generate past examples for interleaved training [De Lange et al., 2021, Scialom et al., 2022]. Recent work has extended CL to large language models, studying instruction tuning across sequential tasks [Wu et al., 2024] and domain-incremental adaptation [Ke et al., 2023]. However, existing CL methods assume a *given* data stream and do not address the problem of *which* experiences should be retained. HAT complements CL by introducing a principled experience selection mechanism upstream of parameter updates.

2.2 Memory-Augmented Neural Networks

Memory-augmented architectures equip neural networks with external memory stores that can be read from and written to [Graves et al., 2014, Sukhbaatar et al., 2015]. More recent work combines retrieval mechanisms with language models [Borgeaud et al., 2022, Zhong et al., 2022], enabling access to a large knowledge base at inference time. While these approaches augment *inference*, they do not leverage memory for *learning*: retrieved information is used contextually but does not update the model’s parameters. In contrast, HAT uses its memory store (the Neocortex) as a curated training set that feeds back into parameter optimisation.

2.3 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) methods prepend retrieved passages to the model’s input context [Lewis et al., 2020, Guu et al., 2020, Izacard et al., 2023]. RAG is effective for knowledge-intensive tasks but inherits the limitations of fixed-context approaches: the model cannot *learn* from

retrieved content, context windows are bounded, and retrieval quality degrades when the knowledge base is noisy. HAT can be seen as going beyond RAG by closing the loop from retrieval to training: high-value interactions—including oracle-provided corrections—are not only used at inference time but are consolidated into the model’s parameters.

2.4 Knowledge Distillation and Self-Improvement

Knowledge distillation transfers knowledge from a teacher model to a student [Hinton et al., 2015, Sanh et al., 2019]. Recent work explores *self-improving* language models that generate their own training data [Huang et al., 2023, Yuan et al., 2024, Chen et al., 2024] or use reinforcement learning from human/AI feedback [Ouyang et al., 2022, Bai et al., 2022]. These methods typically apply distillation or self-play uniformly, without selecting *which* experiences are most informative. HAT’s Oracle consultation mechanism is a form of selective, on-demand distillation: the teacher is queried only when the student model is uncertain or has been corrected by the user, ensuring efficient use of the oracle’s capacity.

2.5 Complementary Learning Systems Theory

Complementary learning systems (CLS) theory posits that the brain uses two interacting memory systems: the hippocampus for rapid encoding of specific episodes and the neocortex for slow extraction of statistical regularities [McClelland et al., 1995, Kumaran et al., 2016]. Sleep replay, particularly during slow-wave sleep, is hypothesised to mediate the transfer from hippocampus to neocortex [Rasch and Born, 2013, Walker, 2017]. Several works in machine learning have drawn on CLS for continual learning [Shin et al., 2017, van de Ven et al., 2020, Arani et al., 2022], but they focus on image classification and do not address the unique challenges of language model self-improvement from interactive experience. HAT directly operationalises the CLS framework for LLMs, with an explicit hippocampal selection stage, a neocortical memory store, and an offline SWS training phase.

3 HAT: Hippocampus-Augmented Transformer

We present **Hippocampus-Augmented Transformer (HAT)**, a memory-centric continual learning approach that enables language models to learn *what to learn* from interaction. HAT is inspired by hippocampal–neocortical memory consolidation in the mammalian brain, where salient experiences are rapidly encoded, selectively retained, and later replayed for gradual integration into long-term cortical memory. Rather than updating a model from all observed interactions, HAT separates online experience acquisition from offline model adaptation: during the *wake phase*, interaction traces are selectively consolidated into a long-term memory; during the *slow-wave sleep* (SWS) phase, selected traces are replayed as training signals to update the model.

As shown in Figure 1, HAT consists of four functional components. The **Cortex** $\mathcal{M}_\theta$ is a lightweight language model that interacts with users. The **Hippocampus Agent** $\mathcal{H}$ abstracts interactions into structured memory traces, estimates their learning value, and determines whether they should be retained. The **Neocortex** $\mathcal{N}$ is a persistent long-term memory that stores selected traces and their replay-ready training forms. The **SWS Trainer** periodically updates the Cortex by replaying curated memories from the Neocortex. An external **Oracle** $\mathcal{O}$ may be queried on demand when the Cortex is uncertain or when user feedback indicates an error.

3.1 Selective Learning Formulation

HAT treats continual adaptation as a selective learning problem over an interaction stream. Let

$$\mathcal{D}^{(t)} = \{x_i\}_{i=1}^{T_t} \quad (1)$$

denote the set of interactions observed during wake cycle t , where each interaction is represented as

$$x_i = (c_i, x_i, y_i, f_i), \quad (2)$$

with context c_i , user query x_i , model response y_i , and optional feedback signal f_i . The goal is not to learn from every interaction, but to identify a subset of experiences that are likely to improve the model under a limited adaptation budget.

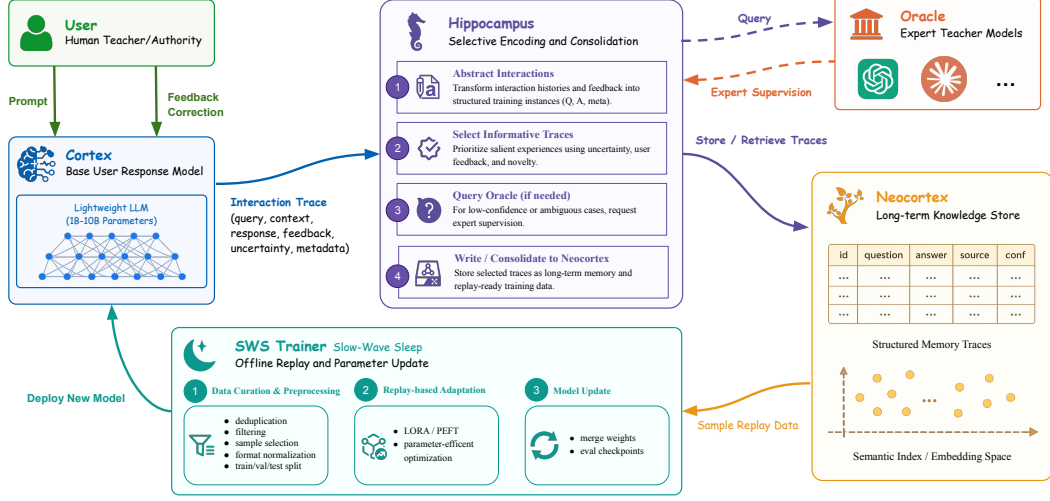


Figure 1: Overview of HAT, a hippocampus-inspired continual learning mechanism that converts selected interaction experiences into reusable training signals. During wake-phase interaction, the Cortex responds to user prompts while the Hippocampus encodes, selects, and optionally augments informative traces with Oracle supervision. The selected traces are consolidated into the Neocortex and replayed during slow-wave sleep (SWS), where the SWS Trainer updates the Cortex through parameter-efficient fine-tuning for continual self-improvement.

132 Ideally, one would select experiences according to their expected learning gain:

$$\mathcal{D}^{*(t)} = \arg \max_{\mathcal{D}' \subseteq \mathcal{D}^{(t)}} \mathbb{E}_{x \sim \mathcal{D}'} [\Delta \mathcal{L}(\theta; x)], \quad (3)$$

133 where $\Delta \mathcal{L}(\theta; x)$ denotes the expected reduction in future loss after adapting the model on experience
 134 x . Since this quantity is generally unobservable at selection time, HAT approximates expected
 135 learning gain using observable proxies: model uncertainty, external feedback, and semantic novelty.
 136 This converts continual learning from indiscriminate data accumulation into a selective memory
 137 consolidation process.

138 3.2 Wake-Sleep Learning Cycle

139 HAT alternates between two complementary phases.

140 **Wake phase.** During the wake phase, the Cortex $\mathcal{M}_\theta$ receives a user query x under context
 141 c and generates a response y . The resulting interaction is passed to the Hippocampus Agent,
 142 which compresses it into a memory trace, evaluates its utility, and determines whether it should be
 143 consolidated into the Neocortex. If the Cortex is uncertain or the user provides corrective feedback,
 144 the Hippocampus Agent may consult the Oracle to obtain additional supervision.

145 **Sleep phase.** During the SWS phase, selected traces stored in the Neocortex are replayed as training
 146 examples. The SWS Trainer updates the Cortex parameters using a curated replay distribution:

$$\theta^{(t+1)} = \text{SWS} \left(\theta^{(t)}, \mathcal{N}^{(t)} \right). \quad (4)$$

147 The updated model then becomes the Cortex for the next wake cycle. Thus, HAT separates the
 148 decision of *what should be learned* from the optimization process that performs the actual learning.

149 3.3 Cortex: Online Interaction Model

150 The Cortex is an autoregressive Transformer language model $\mathcal{M}_\theta$ with parameters θ . It can be
 151 instantiated by any lightweight model suitable for deployment, such as a model in the 0.5–10B

parameter range. Given context c and query x , the Cortex generates a response $y = (y_1, \dots, y_{|y|})$ according to

$$p_\theta(y \mid c, x) = \prod_{j=1}^{|y|} p_\theta(y_j \mid c, x, y_{<j}). \quad (5)$$

In addition to producing a response, the Cortex provides an uncertainty estimate $U(x)$ for each interaction. In practice, $U(x)$ can be computed from predictive entropy, token-level confidence, self-consistency variance, or disagreement between sampled generations. A simple entropy-based estimate is

$$U(x) = \frac{1}{|y|} \sum_{j=1}^{|y|} H(p_\theta(\cdot \mid c, x, y_{<j})), \quad (6)$$

where $H(\cdot)$ denotes entropy. High uncertainty indicates a potential epistemic gap and serves as one signal for selective memory consolidation.

3.4 Hippocampus Agent: Selective Memory Consolidation

The Hippocampus Agent is the central mechanism of HAT. It transforms raw interactions into structured memory traces and determines which traces should be consolidated into the Neocortex. This process consists of three operations: abstraction, selection, and replay construction.

3.4.1 Experience Abstraction

Raw interaction logs are often verbose, redundant, and unsuitable for direct training. The abstraction operation maps an interaction $x = (c, x, y, f)$ into a compact memory trace:

$$m = \mathcal{H}_{\text{abs}}(c, x, y, f). \quad (7)$$

The trace m preserves the learning-relevant content of the episode, including the user intent, the model response, the corrected answer if available, and relevant contextual constraints. This operation is analogous to rapid hippocampal encoding, where an experience is compressed into a retrievable trace rather than stored as an unfiltered sensory stream.

In implementation, $\mathcal{H}_{\text{abs}}$ can be instantiated as a prompting procedure, a small summarization model, or the Cortex itself under an instruction template. A memory trace may be represented as

$$m = (q, a_{\text{cortex}}, a_{\text{target}}, r, \xi), \quad (8)$$

where q is the normalized query, a_{cortex} is the Cortex response, a_{target} is the preferred target answer, r is the rationale or correction signal, and ξ contains timestamp, source, uncertainty, feedback, and novelty statistics.

3.4.2 Selection via Write Policy

Not all traces should be stored or replayed. HAT therefore defines a write policy π_{write} that estimates the consolidation value of each memory trace. Given a trace m , the policy computes

$$\text{score}(m) = \alpha U(m) + \beta F(m) + \gamma N(m), \quad (9)$$

where U , F , and N denote uncertainty, feedback, and novelty, respectively.

Uncertainty. $U(m)$ estimates the extent to which the Cortex is unreliable on the corresponding interaction. High uncertainty suggests that the trace lies near the boundary of the model’s current competence and may yield high learning value.

Feedback. $F(m)$ captures explicit or implicit supervision. For example, user corrections, negative ratings, or contradiction signals can be mapped to a binary or graded value. When an explicit correction is provided, $F(m)$ is assigned high priority to ensure that verified errors are consolidated.

186 **Novelty.** $N(m)$ measures the semantic distance between the current trace and the existing Neocortex
 187 memory. Let $e(m)$ be an embedding of the trace and $\mathcal{N}$ be the current memory store. A practical
 188 novelty score is

$$N(m) = 1 - \max_{m' \in \mathcal{N}} \cos(e(m), e(m')). \quad (10)$$

189 Novel traces are prioritized because they expand coverage of underrepresented regions in the model’s
 190 knowledge space.

191 The final write decision is

$$z(m) = \mathbb{1}[\text{score}(m) > \tau], \quad (11)$$

192 where τ is a tunable consolidation threshold. The coefficients α, β, γ control the relative importance
 193 of epistemic uncertainty, external supervision, and semantic coverage.

194 This write policy can be interpreted as a tractable surrogate for Equation 3: uncertainty approximates
 195 knowledge gaps, feedback identifies externally verified errors, and novelty encourages memory
 196 diversity.

197 3.4.3 Replay Construction

198 Selected memory traces must be converted into training-ready signals. For each retained trace m , the
 199 Replay Builder produces one or more supervised examples:

$$(x^{\text{train}}, y^{\text{train}}) \sim p_{\text{replay}}(\cdot | m) = \mathcal{H}_{\text{replay}}(m). \quad (12)$$

200 The replay distribution can include direct instruction–response pairs, corrected QA pairs, rationale-
 201 augmented examples, or contrastive pairs that distinguish the original erroneous response from the
 202 preferred target.

203 This step is important because memory consolidation in HAT is not merely storage. A retained trace
 204 becomes useful only when it is transformed into a form that can drive gradient-based learning. Thus,
 205 the Hippocampus Agent converts interaction experience into reusable training signals.

206 3.5 Oracle: On-Demand External Supervision

207 The Oracle $\mathcal{O}$ is an external teacher used only when additional supervision is expected to be valuable.
 208 Rather than distilling a large teacher model over an entire corpus, HAT queries the Oracle selectively.
 209 The Oracle is invoked when the uncertainty exceeds a threshold $\tau_{\mathcal{O}}$ or when user feedback indicates
 210 an error:

$$y^{\mathcal{O}} = \mathcal{O}(c, x), \quad \text{if } U(x) > \tau_{\mathcal{O}} \text{ or } F(x) > 0. \quad (13)$$

211 The Oracle response may replace, refine, or supplement the Cortex response in the memory trace:

$$a_{\text{target}} = \text{Merge}(a_{\text{cortex}}, a_{\text{user}}, a_{\mathcal{O}}), \quad (14)$$

212 where a_{user} denotes a user-provided correction when available. This mechanism implements budget-
 213 aware supervision: teacher knowledge is acquired primarily at the frontier of the student’s uncertainty
 214 or error, where it is most likely to improve future behavior.

215 3.6 Neocortex: Long-Term Memory Store

216 The Neocortex $\mathcal{N}$ is a persistent memory store containing selected traces and their replay-ready
 217 training forms. At wake cycle t , the memory is updated as

$$\mathcal{N}^{(t)} = \mathcal{N}^{(t-1)} \cup \{m_i : z(m_i) = 1\}. \quad (15)$$

218 Each memory entry stores both semantic content and training metadata:

$$\mathcal{N} = \{(m_i, \mathcal{R}_i, \text{score}_i, \xi_i)\}_{i=1}^{|\mathcal{N}|}, \quad (16)$$

219 where $\mathcal{R}_i$ denotes replay examples generated from trace m_i . In implementation, the Neocortex can
 220 be realized as a vector-indexed datastore with structured fields, allowing retrieval, deduplication,
 221 pruning, and replay sampling.

222 To keep memory bounded, HAT may prune low-value or redundant traces:

$$\mathcal{N} \leftarrow \text{TopK}(\mathcal{N}, \text{score}(m)), \quad (17)$$

223 or apply diversity-aware sampling to avoid overrepresenting frequent interaction patterns. This
224 prevents indiscriminate accumulation and maintains the Neocortex as a curated long-term learning
225 substrate.

226 3.7 Slow-Wave Sleep Trainer

227 During the SWS phase, HAT updates the Cortex by replaying selected experiences from the Neocortex.
228 Let $\mathcal{B} \subseteq \mathcal{N}$ denote a replay batch sampled from the memory store. The SWS Trainer performs
229 empirical risk minimization over the curated replay distribution:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim p_{\text{replay}}} [\mathcal{L}_{\text{sup}}(\theta; x, y)] + \mathcal{R}(\theta). \quad (18)$$

230 In practice, the supervised loss is the token-level negative log-likelihood:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} \sum_{j=1}^{|y|} \log p_{\theta}(y_j | x, y_{<j}). \quad (19)$$

231 For traces involving Oracle supervision, HAT optionally includes a distillation term:

$$\mathcal{L}_{\text{KD}} = \frac{1}{|\mathcal{B}_{\mathcal{O}}|} \sum_{x \in \mathcal{B}_{\mathcal{O}}} \text{KL}(p_{\mathcal{O}}(\cdot | x) \| p_{\theta}(\cdot | x)), \quad (20)$$

232 where $\mathcal{B}_{\mathcal{O}}$ is the subset of replay examples that contain Oracle-derived targets.

233 To reduce catastrophic forgetting, the SWS objective may include a stability regularizer:

$$\mathcal{L}_{\text{stab}} = \frac{\mu}{2} \sum_i F_i (\theta_i - \theta_i^{\text{old}})^2, \quad (21)$$

234 where F_i denotes the estimated importance of parameter θ_i , such as the diagonal Fisher information
235 as in elastic weight consolidation [Kirkpatrick et al., 2017].

236 The full SWS objective is

$$\mathcal{L}_{\text{SWS}} = \mathcal{L}_{\text{sup}} + \lambda_{\text{KD}} \mathcal{L}_{\text{KD}} + \lambda_{\text{stab}} \mathcal{L}_{\text{stab}}. \quad (22)$$

237 The Cortex parameters are then updated as

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}_{\text{SWS}}(\theta^{(t)}; \mathcal{B}^{(t)}). \quad (23)$$

238 For efficient deployment, HAT can instantiate this update using parameter-efficient fine-tuning
239 methods such as LoRA or adapters, while keeping most model parameters frozen. This makes the
240 SWS phase practical for small or edge-deployed language models.

241 3.8 Iterative Self-Improvement

242 The complete HAT process forms a recurrent wake–sleep learning loop. At cycle t , the current Cortex
243 interacts with users and produces an interaction stream $\mathcal{D}^{(t)}$. The Hippocampus Agent abstracts
244 and filters this stream into selected memory traces. The Neocortex stores these traces as long-term
245 memory. The SWS Trainer then replays the selected traces to produce the next Cortex:

$$\theta^{(t+1)} = \mathcal{A}(\theta^{(t)}, \pi_{\text{write}}(\mathcal{D}^{(t)}, \mathcal{N}^{(t)})), \quad (24)$$

246 where $\mathcal{A}$ denotes the SWS adaptation operator and π_{write} denotes the selective consolidation policy.

247 This loop creates a self-improving process in which each model generation produces new interaction
248 evidence, the Hippocampus Agent decides which experiences are worth retaining, and the SWS phase
249 converts retained experiences into parameter updates. Importantly, HAT does not require the model to
250 store all interactions or continuously update online. Instead, it learns through selective consolidation:
251 wake-phase interactions determine *what* should be learned, while sleep-phase replay determines *how*
252 the model is updated.

253 Algorithm 1 summarizes the full procedure.

Algorithm 1: HAT wake-sleep learning loop

Input: Initial Cortex parameters $\theta^{(0)}$, Neocortex memory $\mathcal{N}^{(0)}$, Oracle $\mathcal{O}$, thresholds $\tau, \tau_{\mathcal{O}}$

```
1 for wake-sleep cycle  $t = 0, 1, 2, \dots$  do
2   Collect wake-phase interactions  $\mathcal{D}^{(t)}$ 
3   for each interaction  $x = (c, x, y, f) \in \mathcal{D}^{(t)}$  do
4     Estimate uncertainty  $U(x)$  from Cortex  $\mathcal{M}_{\theta}$ 
5     Encode memory trace  $m \leftarrow \mathcal{H}_{\text{abs}}(x)$ 
6     if  $U(x) > \tau_{\mathcal{O}}$  or  $F(x) > 0$  then
7       Query Oracle  $y^{\mathcal{O}} \leftarrow \mathcal{O}(c, x)$ 
8       Update trace target using Oracle/user supervision
9     Compute novelty  $N(m)$  against  $\mathcal{N}^{(t)}$ 
10    Compute score  $\text{score}(m) = \alpha U + \beta F + \gamma N$ 
11    if  $\text{score}(m) > \tau$  then
12      Build replay examples  $\mathcal{R} \leftarrow \mathcal{H}_{\text{replay}}(m)$ 
13      Store  $(m, \mathcal{R}, \text{score}, \xi)$  in  $\mathcal{N}^{(t)}$ 
14  Sample replay batch  $\mathcal{B}^{(t)} \subseteq \mathcal{N}^{(t)}$ 
15  Update Cortex via SWS training:  $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \nabla_{\theta} \mathcal{L}_{\text{SWS}}$ 
16  Optionally prune redundant or low-value memories from  $\mathcal{N}^{(t)}$ 
```

254 4 Experiments

255 4.1 Benchmarks and Tasks

256 We evaluate HAT on three families of tasks that probe different aspects of self-improvement:

257 **Knowledge-intensive question answering.** We use Natural Questions (NQ) [Kwiatkowski et al.,
258 2019] and TriviaQA [Joshi et al., 2017] in a *streaming* setup: questions arrive sequentially and the
259 model must answer without access to a retrieval corpus. After each answer, the model may receive a
260 binary correctness signal or an explicit user correction.

261 **Instruction following.** We adopt a subset of the Alpaca-Eval benchmark [Li et al., 2023] and
262 MT-Bench [Zheng et al., 2023], where instructions are presented in temporal batches that gradually
263 increase in complexity. This tests HAT’s ability to internalise and generalise from corrective feedback
264 on instruction-following quality.

265 **Personalisation.** We construct a personalisation benchmark based on LaMP [Salemi et al., 2024],
266 where a simulated user provides preferences and corrections over successive interactions. The model
267 is evaluated on how well it adapts to individual user style and factual preferences over time.

268 4.2 Baselines

269 We compare HAT against the following approaches:

- 270 • **Frozen:** the base Cortex model without any adaptation.
- 271 • **RAG** [Lewis et al., 2020]: retrieval-augmented generation using a dense retriever over the interac-
272 tion history.
- 273 • **Naïve Replay:** continual fine-tuning on all interaction logs without selection.
- 274 • **EWC Replay** [Kirkpatrick et al., 2017]: replay with elastic weight consolidation but without
275 experience curation.
- 276 • **DER++** [Buzzega et al., 2020]: dark experience replay with knowledge distillation, a strong
277 continual learning baseline.
- 278 • **Self-Play** [Chen et al., 2024]: the model generates and filters its own training data via self-
279 consistency.

Table 1: Main results after 10 SWS cycles. We report accuracy (%) on knowledge QA (NQ / TriviaQA, exact match), instruction following (MT-Bench, GPT-4 judge score $\times 10$), and personalisation (LaMP, ROUGE-L $\times 100$). Best in **bold**; second best underlined. [†]Full Distillation uses the oracle for *every* query (upper bound).

Method	Knowledge QA			Instruction Following			Personalisation		
	1.5B	3.8B	8B	1.5B	3.8B	8B	1.5B	3.8B	8B
Frozen	–	–	–	–	–	–	–	–	–
RAG	–	–	–	–	–	–	–	–	–
Naïve Replay	–	–	–	–	–	–	–	–	–
EWC Replay	–	–	–	–	–	–	–	–	–
DER++	–	–	–	–	–	–	–	–	–
Self-Play	–	–	–	–	–	–	–	–	–
Full Distill. [†]	–	–	–	–	–	–	–	–	–
HAT (ours)	–	–	–	–	–	–	–	–	–

280 • **Full Distillation:** fine-tuning the student on the oracle’s responses for *all* queries (upper bound on
281 oracle usage).

282 4.3 Implementation Details

283 **Base models.** We instantiate the Cortex with three model families at different scales: Qwen-2.5-
284 1.5B, Phi-3-mini-3.8B, and LLaMA-3-8B. All models are instruction-tuned variants.

285 **Oracle.** We use GPT-4o as the default oracle $\mathcal{O}$. We also report results with a smaller oracle
286 (LLaMA-3-70B) to study oracle quality sensitivity.

287 **Hippocampus Agent.** The abstraction and replay-builder modules reuse the Cortex itself (with
288 dedicated prompts), making HAT lightweight and self-contained. The selection scoring uses the
289 Cortex’s own predictive entropy for U , cosine similarity in the model’s embedding space for N , and
290 binary user-correction flags for F .

291 **Hyperparameters.** We set $\alpha = 0.4$, $\beta = 0.4$, $\gamma = 0.2$ in Eq. (9) based on validation-set tuning.
292 The write threshold is $\tau = 0.3$. Oracle consultation is triggered when $U > \tau_{\mathcal{O}} = 0.7$ or when $F = 1$.
293 SWS training uses LoRA (rank 16) with AdamW ($\eta = 2 \times 10^{-5}$), batch size 32, and 3 epochs per
294 cycle. The EWC coefficient is $\mu = 0.1$. A SWS cycle is triggered every 500 interactions.

295 **Evaluation protocol.** We report metrics averaged over 3 random seeds. Each experiment simu-
296 lates 5,000 sequential interactions (10 SWS cycles). We measure: (1) **accuracy** (exact match or
297 ROUGE-L depending on task), (2) **error rate** (fraction of incorrect responses), (3) **sample efficiency**
298 (accuracy gain per training example), and (4) **forgetting rate** (performance drop on earlier tasks after
299 adaptation).

300 5 Results

301 5.1 Main Results

302 Table 1 summarises the performance of HAT and all baselines after 5,000 interactions (10 SWS
303 cycles) across three benchmark families and three base model scales.

304 **HAT consistently outperforms non-parametric baselines.**

305 **HAT surpasses continual learning baselines.**

306 **Selective curation closes the gap to full distillation.**

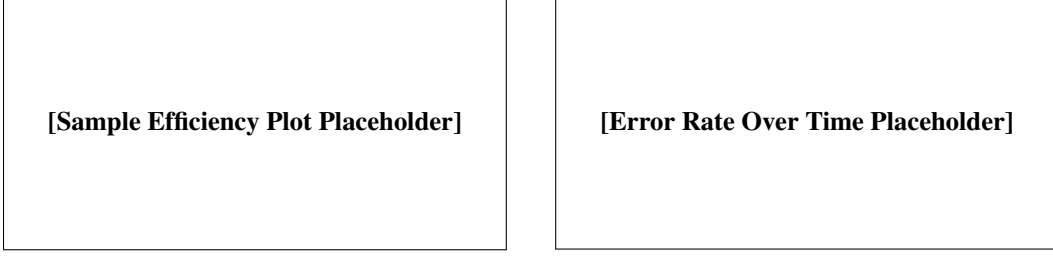


Figure 2: **Left:** Accuracy vs. number of training examples consumed (sample efficiency). **Right:** Error rate over successive SWS cycles. HAT achieves higher accuracy with fewer examples and reduces errors faster than baselines.

Table 2: Impact of oracle budget on HAT performance (Phi-3-3.8B, NQ benchmark). “% Oracle” is the fraction of interactions that trigger an oracle call.

Setting	% Oracle	Accuracy	Error Rate
HAT (no oracle)	0%	–	–
HAT ($\tau_O = 0.9$)	–%	–	–
HAT ($\tau_O = 0.7$)	–%	–	–
HAT ($\tau_O = 0.5$)	–%	–	–
Full Distillation	100%	–	–

307 5.2 Sample Efficiency

308 Figure 2 plots accuracy as a function of the number of training examples consumed.

309 5.3 Oracle Budget Analysis

310 A practical concern is the cost of oracle consultations. Table 2 reports performance and the number
311 of oracle calls made under different oracle-budget constraints.

312 6 Analysis

313 6.1 Ablation Study

314 To understand the contribution of each component, we ablate key modules of HAT using the Phi-3-
315 3.8B Cortex on the NQ benchmark. Results are summarised in Table 3.

316 **Abstraction is critical.**

317 **Selection outperforms random curation.**

318 **All three selection signals contribute.**

319 **EWC prevents forgetting.**

320 6.2 Effect of SWS Cycle Frequency

321 We vary the number of interactions between SWS cycles ($N \in \{100, 250, 500, 1000, 2000\}$) and
322 measure final accuracy and computational cost.

323 6.3 Qualitative Analysis

324 **What does the Hippocampus select?** We analyse the distribution of memory traces written to the
325 Neocortex across interaction types.

Table 3: Ablation study on the NQ benchmark (Phi-3-3.8B). Each row removes or replaces one component. Δ indicates the accuracy change relative to the full HAT system.

Variant	Accuracy	Δ
HAT (full)	—	—
w/o Abstraction	—	—
w/o Selection (random write)	—	—
w/o Novelty signal ($\gamma = 0$)	—	—
w/o Uncertainty signal ($\alpha = 0$)	—	—
w/o Feedback signal ($\beta = 0$)	—	—
w/o Oracle consultation	—	—
w/o EWC regularisation	—	—
w/o Replay Builder (raw traces)	—	—

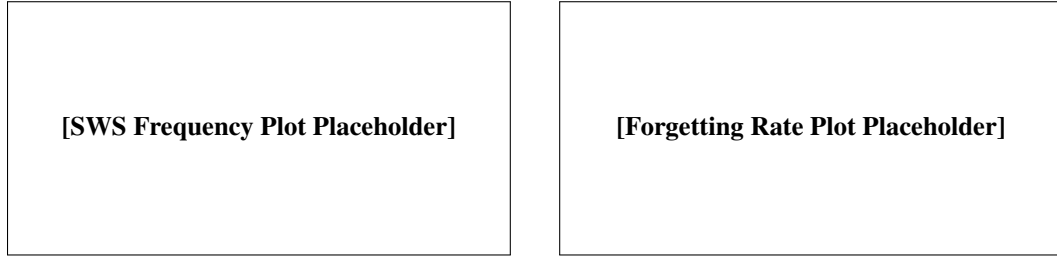


Figure 3: **Left:** Final accuracy as a function of SWS cycle frequency. **Right:** Forgetting rate across SWS cycles for HAT vs. baselines.

326 **Memory evolution over time.** We visualise the Neocortex content across SWS cycles using t-SNE
 327 embeddings.

328 **Case study: error correction propagation.** We trace a specific user correction through the HAT
 329 pipeline—from raw interaction to abstracted trace, selection, replay construction, and its effect on
 330 model behaviour after SWS training.

331 6.4 Scaling Behaviour

332 We examine how HAT’s gains vary with Cortex model size (1.5B, 3.8B, 8B).

333 7 Conclusion

334 We introduced **HAT (Hippocampus-Augmented Transformer)**, a memory-centric framework that
 335 enables lightweight language models to self-improve by learning *what to learn* from interactive expe-
 336 rience. Drawing on complementary learning systems theory, HAT decouples experience acquisition
 337 (online) from parameter optimisation (offline) through three key mechanisms: abstraction of raw
 338 interactions into structured memory traces, selective curation based on uncertainty, feedback, and
 339 novelty, and offline replay-driven fine-tuning during slow-wave sleep phases. An on-demand oracle
 340 consultation mechanism provides efficient, targeted knowledge distillation only where the model is
 341 most uncertain.

342 Experiments across knowledge-intensive QA, instruction following, and personalisation benchmarks
 343 demonstrate that HAT consistently outperforms strong fine-tuning and retrieval-augmented baselines
 344 in accuracy, sample efficiency, and resistance to catastrophic forgetting, while using a fraction of the
 345 oracle budget required by full distillation.

346 **Limitations and future work.** Several directions remain open. First, the current selection heuristic
 347 (Eq. 9) uses a linear combination of fixed signals; learning the write policy end-to-end via meta-
 348 learning is a promising extension. Second, our evaluation simulates user interactions from existing
 349 benchmarks; deploying HAT with real users in a longitudinal study would provide stronger evidence

of its practical impact. Third, while HAT targets lightweight models (0.5–10 B), scaling the framework to larger models and investigating the interplay between model capacity and memory curation is of both theoretical and practical interest. Finally, privacy considerations arise when storing user interactions in the Neocortex; integrating differential privacy or federated learning into the SWS training phase is an important direction for responsible deployment.

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456 A Algorithm Pseudocode

Algorithm 2: HAT — Hippocampus-Augmented Transformer Self-Improvement Loop

Input : Base model $\mathcal{M}_\theta$ (Cortex), Hippocampus Agent $\mathcal{H}$, Oracle $\mathcal{O}$, empty Neocortex $\mathcal{N} \leftarrow \emptyset$
Input : SWS interval N , write threshold τ , oracle threshold $\tau_{\mathcal{O}}$, selection weights α, β, γ

```

1  $t \leftarrow 0$ 
2 for each user interaction  $(c, x)$  do
3    $y \leftarrow \mathcal{M}_\theta(c, x)$  // Cortex generates response
4    $U \leftarrow H(p_\theta(y | c, x))$  // Estimate uncertainty
5   Observe user feedback  $F$  (correction signal, if any)
   // -- Hippocampus Agent --
6    $m \leftarrow \mathcal{H}_{\text{abs}}(c, x, y)$  // Abstraction
7   if  $U > \tau_{\mathcal{O}}$  or  $F = 1$  then
8      $y^{\mathcal{O}} \leftarrow \mathcal{O}(c, x)$  // Consult Oracle
9     Update  $m$  with oracle response  $y^{\mathcal{O}}$ 
10  Compute novelty:  $N \leftarrow \text{NoveltyScore}(m, \mathcal{N})$ 
11  Compute selection score:  $s \leftarrow \alpha U + \beta F + \gamma N$ 
12  if  $s > \tau$  then
13     $(x^{\text{train}}, y^{\text{train}}) \leftarrow \mathcal{H}_{\text{replay}}(m)$  // Replay Builder
14     $\mathcal{N} \leftarrow \mathcal{N} \cup \{(x^{\text{train}}, y^{\text{train}}, s)\}$  // Write to Neocortex
15   $t \leftarrow t + 1$ 
   // -- Slow-Wave Sleep --
16  if  $t \bmod N = 0$  then
17    Sample replay batch  $\mathcal{B} \sim \mathcal{N}$  (priority sampling by  $s$ )
18     $\theta \leftarrow \theta - \eta \nabla_\theta [\mathcal{L}_{\text{SWS}}(\theta; \mathcal{B}) + \mathcal{L}_{\text{reg}}(\theta)]$  // SWS update
19    Update Fisher information  $F$  for EWC
20 return Updated model  $\mathcal{M}_\theta$ 

```

Table 4: Hyperparameter sensitivity analysis on the NQ benchmark (Phi-3-3.8B).

Hyperparameter	Range Tested	Best Value	Sensitivity
Write threshold τ	{0.1, 0.2, 0.3, 0.5, 0.7}	0.3	Medium
Oracle threshold $\tau_{\mathcal{O}}$	{0.5, 0.6, 0.7, 0.8, 0.9}	0.7	Low
Uncertainty weight α	{0.2, 0.3, 0.4, 0.5, 0.6}	0.4	Medium
Feedback weight β	{0.2, 0.3, 0.4, 0.5, 0.6}	0.4	Low
Novelty weight γ	{0.1, 0.2, 0.3, 0.4}	0.2	Low
LoRA rank	{4, 8, 16, 32}	16	Low
SWS learning rate η	$\{1, 2, 5\} \times 10^{-5}$	2×10^{-5}	Medium
EWC coefficient μ	{0.01, 0.05, 0.1, 0.5, 1.0}	0.1	Medium
SWS interval N	{100, 250, 500, 1000, 2000}	500	Low

Table 5: Computational overhead of HAT components (Phi-3-3.8B on a single NVIDIA A100 GPU).

Component	Time per Interaction	Time per SWS Cycle
Cortex inference	— ms	—
Hippocampus (abstraction)	— ms	—
Hippocampus (selection)	— ms	—
Oracle consultation	— ms	—
SWS training (3 epochs)	—	— min
Total overhead	— ms	— min

B Hyperparameter Sensitivity

C Additional Experimental Results

D Prompts Used by the Hippocampus Agent

E Computational Cost Analysis

F Broader Impact Statement

HAT enables on-device language models to improve from user interactions, which raises considerations in both positive and negative directions.

Positive impacts. HAT democratises model adaptation by allowing lightweight models on personal devices to improve without requiring cloud-based retraining. This can reduce reliance on centralised AI services and give users more control over their AI assistants. The selective memory mechanism also implies that not all data is stored, which can be beneficial for privacy.

Potential risks. Storing user interactions, even in abstracted form, poses privacy risks. If the Neocortex memory is compromised, sensitive information could be exposed. Additionally, a model that learns from user corrections could amplify biases if the corrections themselves are biased. We recommend deploying HAT with differential privacy mechanisms and content filtering to mitigate these risks.

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717 strategies (e.g., gated release of models, providing defenses in addition to attacks,
718 mechanisms for monitoring misuse, mechanisms to monitor how a system learns from
719 feedback over time, improving the efficiency and accessibility of ML).

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721 Question: Does the paper describe safeguards that have been put in place for responsible
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723 image generators, or scraped datasets)?

724 Answer: **[TODO]**

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